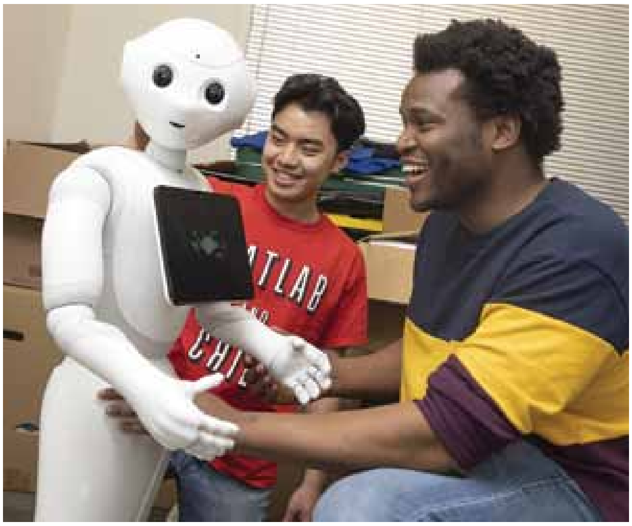
Driving simulation checks how intentional robotic deception affects trust

Kantwon Rogers, a Ph.D. student in the College of Computing, and Reiden Webber, a second-year computer science undergraduate at Georgia Tech, have designed a driving



Kantwon Rogers (right), a Ph.D. student in the College of Computing and lead author on the study, and Reiden Webber, a second-year undergraduate student in computer science (Credit: Georgia Institute of Technology)

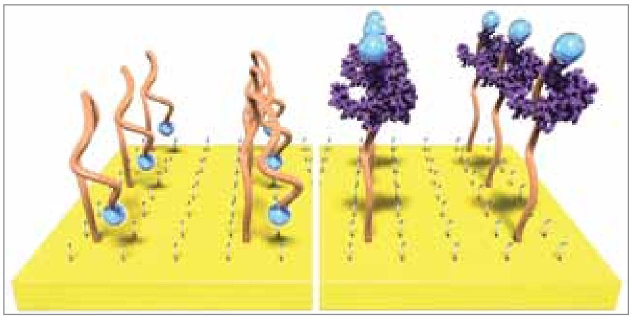
simulation to investigate how intentional robot deception affects trust. Researchers designed a driving simulation to study how people interact with AI under high-stake, time-sensitive situations. They recruited 341 online and 20 in-person participants who completed a trust measurement survey beforehand to gauge their preconceived notions about AI behaviour. Participants completed a driving simulation with a robot assistant instructing them to obey the speed limit and reach a checkpoint. After completing the task, participants were asked about the robot's behaviour and their trust was measured again. An additional 100 online participants completed the simulation without any robot assistant.

New AI architecture likely to solve complex reasoning tasks efficiently

Researchers at IBM Research Zürich and ETH Zürich have created a new architecture that combines two of the most renowned artificial intelligence approaches, namely deep neural networks and vector-symbolic models. Researchers developed two key enablers of architecture. The first is new neural network training for flexible representation learning and the second is a VSA transformation method to simplify probability computations with algebraic operations. The architecture outperformed other models in solving RAVEN's matrices, achieving record accuracies of 87.7% on RAVEN and 88.1% on I-RAVEN datasets. NVSA performs extensive probabilistic calculations in a single vector operation, enabling it to solve abstract reasoning problems faster and more accurately than other AI approaches. This new architecture shows promise for efficiently solving complex reasoning tasks.

A new transistor based sensor amplifies weak biochemcal signals

Researchers from Northwestern University have developed a new technology that makes it easier to eavesdrop on our body's inner conversations. This new method makes signals easier to detect without complex and bulky electronics and even boosts signals by more than 1,000 times. Rivnay's team added an amplifying component to a traditional



Single strands of engineered DNA, called aptamers, bind to a target and then fold to trigger an electrochemical signal (Credit: Jonathan Rivnay/ Northwestern University)

sensor and developed a new transistor based sensor with a built-in reference electrode to sense and amplify weak biochemical signals. To validate, researchers measured cytokine levels near a wound, using the new technology to amplify the signal 3-4 times more than traditional electrode based aptamer sensing methods. The concept is intended to be implemented into implantable biosensors or wearable devices that can sense and respond to a problem.

Analysis of neural networks proves they can be designed to be 'optimal'

MIT researchers conducted an analysis of neural networks and proved that they can be designed so they are 'optimal,' meaning they minimise the probability of misclassifying borrowers or patients into the wrong category when the networks are given a lot of labeled training data. Researchers identified three ways to classify inputs in this network. One method relies on majority, another selects